**0/8** Questions Answered

## Project initial implementation - data team member

### Q1 Joint responsibilities

**1 Point**

Please upload:

A pair of JSON files representing

- one input sample (with real representative values)
- and one model output (again, with real representative values).

(Training, serving, and data team members must agree on this item together; training and inference team members ingest an input like the sample and produce an output like the sample, data team member produces something like the input sample in online and offline workflows.)

If your project involves more than one model, you will have one JSON pair per model. In that case, please name each JSON clearly to indicate which model it "belongs" to.

Input JSON sample(s):

Please select file(s) Select file(s)

Output JSON samples(s):

Please select file(s) Select file(s)

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(4-person team only) If you are part of a four-person team, also upload

- a table enumerating all the containers involved in each role, with links to their Dockerfiles/Docker Compose files, and a link to the equivalent K8S manifest for each one.

The object is to show that each of the individual role-owned systems will be supported by the DevOps/Platform role, although at this stage team members are working independently.

> **Upload in PDF format.**

Please select file(s) Select file(s)

Save Answer

## Q2 Data contributions
**9 Points**

### Q2.1 High-level data design document
**1 Point**

Upload your high-level data design document. You should enumerate the data repositories (databases, lakehouses, object storage buckets, etc.) that will be used, and for each, identify:

- What data is stored there, and the data schema
- What services and processes write/update the data, and when
- How it is versioned (data that will be used to train models must be versioned, along with enough information to track how it entered the system and how it was transformed within the system)
  and you should also have one or more diagrams that show the data flow.

Note: this is not a planning document, it reflects what you have already implemented! The remaining items in this submission (artifacts + videos) should show how you implement what you describe in the design document.

**Upload in PDF format.**

Please select file(s) Select file(s)

Save Answer

### Q2.2 Live object storage bucket
**1 Point**

Provide a link to your object storage bucket on Chameleon, browseable by course staff, with data as illustrated by your data design document.

(If you have multiple buckets, e.g. one at CHI@TACC and one at CHI@UC, list them all.)

Save Answer

### Q2.3 Artifacts + video: External data ingestion
**1.75 Points**

You should have implemented a reproducible pipeline that ingests external data into Chameleon object storage, and executes whatever transformation is necessary to make it ready for training. If the data is small (less than 5GB), as part of the pipeline, you should also expand the data following best practices for synthetic data generation discussed in the lecture.

Upload artifacts (Dockerfiles, Python scripts, DAGs, etc.) for this pipeline:

Please select file(s) Select file(s)

and a sped-up demo video of this pipeline. (Video should demonstrate everything from pipeline launch to external confirmation that it worked.)

Please select file(s) Select file(s)

(You are not required to use a workflow orchestrator at this stage; pipelines can be a `make` target, one-shot jobs in a Docker compose, or run with a sequence of Python scripts. But, all components must be runnable non-interactively from the artifacts.)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

**Q2.4 Artifacts + video: Data generator**
**1.75 Points**

You should have implemented a data generator that hits the (hypothetical) service endpoints with real or synthetic data (following our best practices for synthetic data generation, as discussed in the lecture).

Upload artifacts (Dockerfiles, Python scripts, DAGs, etc.) for this pipeline:

Please select file(s) Select file(s)

and a sped-up demo video of this pipeline. (Video should demonstrate launch + a few minutes of runtime.)

Please select file(s) Select file(s)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

**Q2.5 Artifacts + video: Online feature computation path**
**1.75 Points**

You should have implemented an online feature computation path for real time inference (does not have to be fully integrated with the open source service, but needs to be integrate-able...)

Upload artifacts (Dockerfiles, Python scripts, DAGs, etc.) for this path:

Please select file(s) Select file(s)

and a sped-up demo video of it. (Video should demonstrate at least one end-to-end example.)

Please select file(s) Select file(s)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

**Q2.6 Artifacts + video: Batch jobs**
**1.75 Points**

You should have implemented  batch pipeline that compiles versioned training and evaluation data sets from "production" data, with well-justified candidate selection and avoiding data leakage.

Upload artifacts (Dockerfiles, Python scripts, DAGs, etc.) for this pipeline:

Please select file(s) Select file(s)

and a sped-up demo video of this pipeline. (Video should demonstrate everything from pipeline launch to external confirmation that it worked.)

Please select file(s) Select file(s)

(You are not required to use a workflow orchestrator at this stage; pipelines can be a `make` target, one-shot jobs in a Docker compose, or run with a sequence of Python scripts. But, all components must be runnable non-interactively from the artifacts.)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

## Q3 Bonus

**0 Points**

If you are attempting the extra credit:

Integrate a data framework not used in the lab assignments, that substantially improves your data design (i.e. swapping MariaDB in place of PostgreSQL doesn't count). For example: implement a data transformation layer with `dbt`, add data quality checks with Soda, use a vector database like Qdrant, integrate a Feast feature store, implement distributed computation of features with Spark, add a DataHub data catalog. You must justify why it improves your design, using a concrete example that is realistic in the context of your proposed service.

Submit a PDF document explaining in detail, including the concrete realistic example, and also any artifacts (e.g. Dockerfiles, Python scripts) that go along with it.

Please select file(s) Select file(s)

Note: There is no credit for systems that are not running on Chameleon Cloud. Code or configuration that has not been executed in the target environment, or that only runs locally, does not count. You should *only* submit items that you have implemented and executed on Chameleon.

Save Answer

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Save All Answers

Submit & View Submission